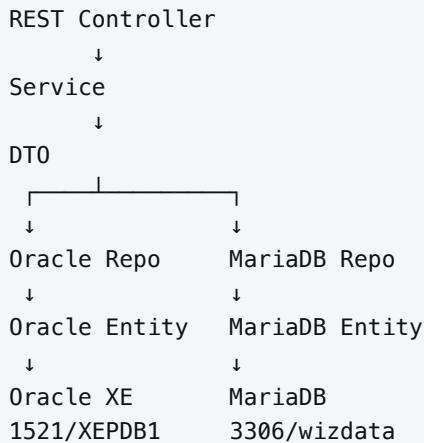


Spring Boot Java + Kotlin Oracle + MariaDB Multi-Database 완성 교재

OracleConfig + MariaDbConfig · Entity · Repository · DTO · Service · Controller · CRUD

1. 전체 아키텍처



text

멀티 DB에서는 Oracle과 MariaDB에 각각 DataSource, EntityManagerFactory, TransactionManager를 구성하고 Repository 패키지를 분리하는 것이 핵심임.

2. 권장 프로젝트 구조

```

src/main/java(or kotlin)/com/example/multidb/
├── MultiDbApplication
├── config/
│   ├── OracleConfig
│   └── MariaDbConfig
├── oracle/
│   ├── entity/OracleMember
│   └── repository/OracleMemberRepository
├── mariadb/
│   ├── entity/MariaMember
│   └── repository/MariaMemberRepository
├── dto/MemberDto
├── service/MemberService
└── controller/MemberController
  
```

text

3. Maven 의존성

```
<dependencies>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
  </dependency>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
  </dependency>
  <dependency>
    <groupId>com.oracle.database.jdbc</groupId>
    <artifactId>ojdbc11</artifactId>
    <scope>runtime</scope>
  </dependency>
  <dependency>
    <groupId>org.mariadb.jdbc</groupId>
    <artifactId>mariadb-java-client</artifactId>
    <scope>runtime</scope>
  </dependency>
</dependencies>
```

xml

Kotlin 프로젝트는 kotlin-reflect, kotlin-stdlib-jdk8 및 Spring/JPA 관련 Kotlin 플러그인을 추가함.

4. application.yml

```
app:
  datasource:
    oracle:
      jdbc-url: jdbc:oracle:thin:@localhost:1521/XEPDB1
      username: devuser
      password: oracle_password
      driver-class-name: oracle.jdbc.OracleDriver

    mariadb:
      jdbc-url: jdbc:mariadb://localhost:3306/wizdata
      username: devuser
      password: mariadb_password
      driver-class-name: org.mariadb.jdbc.Driver

spring:
  jpa:
    open-in-view: false
    properties:
      hibernate:
        format_sql: true
```

yaml

5. Oracle + MariaDB 접속 비교 10개

#	항목	Oracle	MariaDB
1	JDBC URL	jdbc:oracle:thin:@localhost:1521/XEPDB1	jdbc:mariadb://localhost:3306/wizdata
2	기본 포트	1521	3306

#	항목	Oracle	MariaDB
3	Driver	oracle.jdbc.OracleDriver	org.mariadb.jdbc.Driver
4	서비스/DB	XEPDB1	wizdata
5	DataSource Bean	oracleDataSource	mariaDataSource
6	EMF	oracleEntityManagerFactory	mariaEntityManagerFactory
7	Transaction	oracleTransactionManager	mariaTransactionManager
8	Entity 패키지	oracle.entity	mariadb.entity
9	Repository	oracle.repository	mariadb.repository
10	JPA 구현	Hibernate/JPA	Hibernate/JPA

6. Java OracleConfig

```

@Configuration
@EnableTransactionManagement
@EnableJpaRepositories(
    basePackages = "com.example.multidb.oracle.repository",
    entityManagerFactoryRef = "oracleEntityManagerFactory",
    transactionManagerRef = "oracleTransactionManager"
)
public class OracleConfig {

    @Primary
    @Bean
    @ConfigurationProperties("app.datasource.oracle")
    public DataSource oracleDataSource() {
        return DataSourceBuilder.create().build();
    }

    @Primary
    @Bean
    public LocalContainerEntityManagerFactoryBean oracleEntityManagerFactory(
        EntityManagerFactoryBuilder builder) {
        return builder
            .dataSource(oracleDataSource())
            .packages("com.example.multidb.oracle.entity")
            .persistenceUnit("oracle")
            .build();
    }

    @Primary
    @Bean
    public PlatformTransactionManager oracleTransactionManager(
        @Qualifier("oracleEntityManagerFactory")
        EntityManagerFactory emf) {
        return new JpaTransactionManager(emf);
    }
}

```

java

7. Java MariaDbConfig

java

```
@Configuration
@EnableTransactionManagement
@EnableJpaRepositories(
    basePackages = "com.example.multidb.mariadb.repository",
    entityManagerFactoryRef = "mariaEntityManagerFactory",
    transactionManagerRef = "mariaTransactionManager"
)
public class MariaDbConfig {

    @Bean
    @ConfigurationProperties("app.datasource.mariadb")
    public DataSource mariaDataSource() {
        return DataSourceBuilder.create().build();
    }

    @Bean
    public LocalContainerEntityManagerFactoryBean mariaEntityManagerFactory(
        EntityManagerFactoryBuilder builder,
        @Qualifier("mariaDataSource") DataSource dataSource) {
        return builder
            .dataSource(dataSource)
            .packages("com.example.multidb.mariadb.entity")
            .persistenceUnit("mariadb")
            .build();
    }

    @Bean
    public PlatformTransactionManager mariaTransactionManager(
        @Qualifier("mariaEntityManagerFactory")
        EntityManagerFactory emf) {
        return new JpaTransactionManager(emf);
    }
}
```

8. Kotlin OracleConfig

kotlin

```
@Configuration
@EnableTransactionManagement
@EnableJpaRepositories(
    basePackages = ["com.example.multidb.oracle.repository"],
    entityManagerFactoryRef = "oracleEntityManagerFactory",
    transactionManagerRef = "oracleTransactionManager"
)
class OracleConfig {

    @Primary
    @Bean
    @ConfigurationProperties("app.datasource.oracle")
    fun oracleDataSource(): DataSource =
        DataSourceBuilder.create().build()

    @Primary
    @Bean
    fun oracleEntityManagerFactory(
        builder: EntityManagerFactoryBuilder
    ): LocalContainerEntityManagerFactoryBean =
        builder
            .dataSource(oracleDataSource())
            .packages("com.example.multidb.oracle.entity")
            .persistenceUnit("oracle")
            .build()

    @Primary
    @Bean
    fun oracleTransactionManager(
        @Qualifier("oracleEntityManagerFactory") emf: EntityManagerFactory
    ): PlatformTransactionManager =
        JpaTransactionManager(emf)
}
```

9. Kotlin MariaDbConfig

```
@Configuration
@EnableTransactionManagement
@EnableJpaRepositories(
    basePackages = ["com.example.multidb.mariadb.repository"],
    entityManagerFactoryRef = "mariaEntityManagerFactory",
    transactionManagerRef = "mariaTransactionManager"
)
class MariaDbConfig {

    @Bean
    @ConfigurationProperties("app.datasource.mariadb")
    fun mariaDataSource(): DataSource =
        DataSourceBuilder.create().build()

    @Bean
    fun mariaEntityManagerFactory(
        builder: EntityManagerFactoryBuilder,
        @Qualifier("mariaDataSource") dataSource: DataSource
    ): LocalContainerEntityManagerFactoryBean =
        builder
            .dataSource(dataSource)
            .packages("com.example.multidb.mariadb.entity")
            .persistenceUnit("mariadb")
            .build()

    @Bean
    fun mariaTransactionManager(
        @Qualifier("mariaEntityManagerFactory") emf: EntityManagerFactory
    ): PlatformTransactionManager =
        JpaTransactionManager(emf)
}
```

10. Oracle + MariaDB 테이블 생성 SQL 10개 비교

테이블	Oracle 핵심 SQL	MariaDB 핵심 SQL
MEMBER	ID NUMBER GENERATED BY DEFAULT AS IDENTITY, NAME VARCHAR2(100)	id BIGINT AUTO_INCREMENT, name VARCHAR(100)
PRODUCT	PRICE NUMBER(15,2)	price DECIMAL(15,2)
BOARD	CONTENT CLOB	content LONGTEXT
EMPLOYEE	SALARY NUMBER(12,2)	salary DECIMAL(12,2)
CUSTOMER	PHONE VARCHAR2(30)	phone VARCHAR(30)
ORDERS	CREATED_AT DATE DEFAULT SYSDATE	created_at DATETIME DEFAULT CURRENT_TIMESTAMP
PAYMENT	STATUS VARCHAR2(20)	status VARCHAR(20)
UPLOAD_FILE	FILE_PATH VARCHAR2(500)	file_path VARCHAR(500)
CATEGORY	NAME VARCHAR2(100)	name VARCHAR(100)
ACCESS_LOG	MESSAGE VARCHAR2(1000)	message VARCHAR(1000)

11. 대표 MEMBER 테이블 전체 SQL

Oracle

```
CREATE TABLE MEMBER (  
  ID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  NAME VARCHAR2(100) NOT NULL,  
  EMAIL VARCHAR2(200),  
  AGE NUMBER,  
  CREATED_AT DATE DEFAULT SYSDATE  
);
```

sql

MariaDB

```
CREATE TABLE member (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(200),  
  age INT,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

sql

12. Java Entity

```
// OracleMember.java  
@Entity  
@Table(name="MEMBER")  
public class OracleMember {  
  @Id  
  @GeneratedValue(strategy=GenerationType.IDENTITY)  
  private Long id;  
  private String name;  
  private String email;  
  private Integer age;  
  
  public OracleMember() {}  
  // getters/setters  
}  
  
// MariaMember.java  
@Entity  
@Table(name="member")  
public class MariaMember {  
  @Id  
  @GeneratedValue(strategy=GenerationType.IDENTITY)  
  private Long id;  
  private String name;  
  private String email;  
  private Integer age;  
  
  public MariaMember() {}  
  // getters/setters  
}
```

java

13. Kotlin Entity

```
@Entity
@Table(name="MEMBER")
class OracleMember(
    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    var id: Long? = null,
    var name: String = "",
    var email: String? = null,
    var age: Int? = null
)

@Entity
@Table(name="member")
class MariaMember(
    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    var id: Long? = null,
    var name: String = "",
    var email: String? = null,
    var age: Int? = null
)
```

kotlin

14. Java Repository + DTO

```
public interface OracleMemberRepository
    extends JpaRepository<OracleMember, Long> {
    List<OracleMember> findByNameContaining(String name);
}

public interface MariaMemberRepository
    extends JpaRepository<MariaMember, Long> {
    List<MariaMember> findByNameContaining(String name);
}

public record MemberDto(
    Long id,
    String name,
    String email,
    Integer age
) {}
```

java

15. Kotlin Repository + DTO

```
interface OracleMemberRepository :  
    JpaRepository<OracleMember, Long> {  
    fun findByNameContaining(name: String): List<OracleMember>  
}  
  
interface MariaMemberRepository :  
    JpaRepository<MariaMember, Long> {  
    fun findByNameContaining(name: String): List<MariaMember>  
}  
  
data class MemberDto(  
    val id: Long?,  
    val name: String,  
    val email: String?,  
    val age: Int?  
)
```

kotlin

16. JPA Entity → Repository → DTO 예제 10개 비교

#	기능	Java	Kotlin
1	전체 조회	repository.findAll()	repository.findAll()
2	ID 조회	repository.findById(1L)	repository.findById(1L)
3	저장	repository.save(member)	repository.save(member)
4	삭제	repository.deleteById(1L)	repository.deleteById(1L)
5	개수	repository.count()	repository.count()
6	이름검색	List<Member> findByName(String name)	fun findByName(name:String): List<Member>
7	이메일	Optional<Member> findByEmail(String email)	fun findByEmail(email:String): Member?
8	나이이상	findByAgeGreaterThanOrEqualTo(Integer age)	findByAgeGreaterThanOrEqualTo(age:Int)
9	포함검색	findByNameContaining(String name)	findByNameContaining(name:String)
10	정렬	findAll(Sort.by(DESC,"id"))	findAll(Sort.by(DESC,"id"))

17. Java Service - 두 DB 동시 사용

```
@Service
public class MemberService {

    private final OracleMemberRepository oracleRepository;
    private final MariaMemberRepository mariaRepository;

    public MemberService(
        OracleMemberRepository oracleRepository,
        MariaMemberRepository mariaRepository) {
        this.oracleRepository = oracleRepository;
        this.mariaRepository = mariaRepository;
    }

    @Transactional("oracleTransactionManager")
    public OracleMember saveOracle(OracleMember member) {
        return oracleRepository.save(member);
    }

    @Transactional("mariaTransactionManager")
    public MariaMember saveMaria(MariaMember member) {
        return mariaRepository.save(member);
    }

    public List<MemberDto> oracleMembers() {
        return oracleRepository.findAll().stream()
            .map(m -> new MemberDto(
                m.getId(), m.getName(), m.getEmail(), m.getAge()))
            .toList();
    }

    public List<MemberDto> mariaMembers() {
        return mariaRepository.findAll().stream()
            .map(m -> new MemberDto(
                m.getId(), m.getName(), m.getEmail(), m.getAge()))
            .toList();
    }
}
```

java

18. Kotlin Service - 두 DB 동시 사용

```
@Service
class MemberService(
    private val oracleRepository: OracleMemberRepository,
    private val mariaRepository: MariaMemberRepository
) {
    @Transactional("oracleTransactionManager")
    fun saveOracle(member: OracleMember) =
        oracleRepository.save(member)

    @Transactional("mariaTransactionManager")
    fun saveMaria(member: MariaMember) =
        mariaRepository.save(member)

    fun oracleMembers(): List<MemberDto> =
        oracleRepository.findAll().map {
            MemberDto(it.id, it.name, it.email, it.age)
        }

    fun mariaMembers(): List<MemberDto> =
        mariaRepository.findAll().map {
            MemberDto(it.id, it.name, it.email, it.age)
        }
}
```

kotlin

19. Java REST Controller

```
@RestController
@RequestMapping("/api/members")
public class MemberController {

    private final MemberService service;

    public MemberController(MemberService service) {
        this.service = service;
    }

    @GetMapping("/oracle")
    public List<MemberDto> oracle() {
        return service.oracleMembers();
    }

    @GetMapping("/mariadb")
    public List<MemberDto> mariadb() {
        return service.mariaMembers();
    }
}
```

java

20. Kotlin REST Controller

```
@RestController
@RequestMapping("/api/members")
class MemberController(
    private val service: MemberService
) {
    @GetMapping("/oracle")
    fun oracle() = service.oracleMembers()

    @GetMapping("/mariadb")
    fun mariadb() = service.mariaMembers()
}
```

kotlin

21. Java ↔ Kotlin 핵심 비교

항목	Java	Kotlin
DTO	record/class	data class
Repository	extends JpaRepository	: JpaRepository
생성자 주입	생성자 작성	주생성자로 간단히 처리
Null	Optional/@Nullable 등	타입에 ? 표시
Entity	getter/setter 필요	프로퍼티 중심
Service	코드량 상대적으로 많음	간결함
Transaction	@Transactional("...")	@Transactional("...")
JPA 원리	동일	동일
Oracle/MariaDB 분리	Config/Package 분리	Config/Package 분리
추천	기존 Java 프로젝트	신규 간결한 Spring 프로젝트

22. 실행 및 테스트 순서

- 1) Oracle Docker 1521 및 MariaDB Docker 3306이 실행 중인지 확인함.
- 2) Oracle XEPDB1에 devuser 계정과 MEMBER 테이블을 준비함.
- 3) MariaDB wizdata에 devuser 계정과 member 테이블을 준비함.
- 4) application.yml의 IP, 사용자, 비밀번호를 실제 서버 환경에 맞춤.
- 5) Spring Boot를 실행함.
- 6) GET /api/members/oracle 로 Oracle 데이터를 확인함.
- 7) GET /api/members/mariadb 로 MariaDB 데이터를 확인함.
- 8) 저장 메서드에서는 사용할 DB에 맞는 TransactionManager를 명시함.

23. 실무 핵심 주의사항

- 멀티 DataSource에서는 @Primary를 한쪽(본 교재에서는 Oracle)에 명확히 지정함.

- Oracle Repository와 MariaDB Repository의 basePackages가 겹치지 않게 분리함.
- 각 EntityManagerFactory의 packages()가 해당 DB Entity 패키지만 스캔하도록 설정함.
- 한 메서드에서 Oracle과 MariaDB를 동시에 수정하는 경우 단일 로컬 @Transactional만으로 두 DB 전체의 원자성을 자동 보장하지 않음. 분산 트랜잭션이 필요한지는 별도로 설계해야 함.
- 운영 비밀번호를 소스코드나 Git 저장소에 직접 넣지 말고 환경변수/Secret 방식을 권장함.
- Oracle과 MariaDB의 IDENTITY, 날짜형, CLOB/LONGTEXT 등 SQL 차이를 확인함.

데이터 엔지니어링 & RAG 파이프라인 · 부록 — 서버 주소(SERVER_IP)·계정(devuser)·비밀번호(CHANGE_ME...)는 자리표시입니다. 본인 환경의 값으로 바꿔 쓰세요.